

Instructions



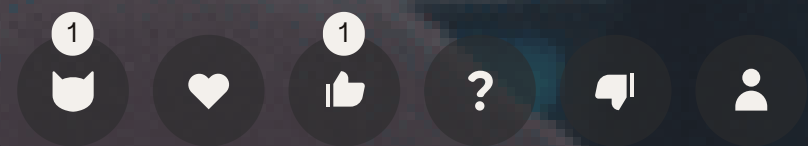
How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

should we make a webpage
no problems everything is good
no nope yes
hello a little bit tired in progress
please explain the homework vse potihonku tiho tiho

Today's agenda

- Total Recall
- Database Layer Design
- Discuss the task

Total Recall



What do you remember from the last session?

mvp mvc mvvp mvp
ggwp mvc jpa
nvm mvvm mvm mvvp

What is the difference between Architectural pattern and Design patterns?

level of concerns!

scale

xd

entire system and
module

the high-level structure of
an entire
application/system, while a
design pattern solves a
smaller software design
problem.

Architectural patterns shape
the big picture while design
patterns solve smaller, recurring
implementation problems
within that structure

level of the problem they
solve

An architectural pattern
shapes the whole system; a
design pattern solves a
specific coding problem

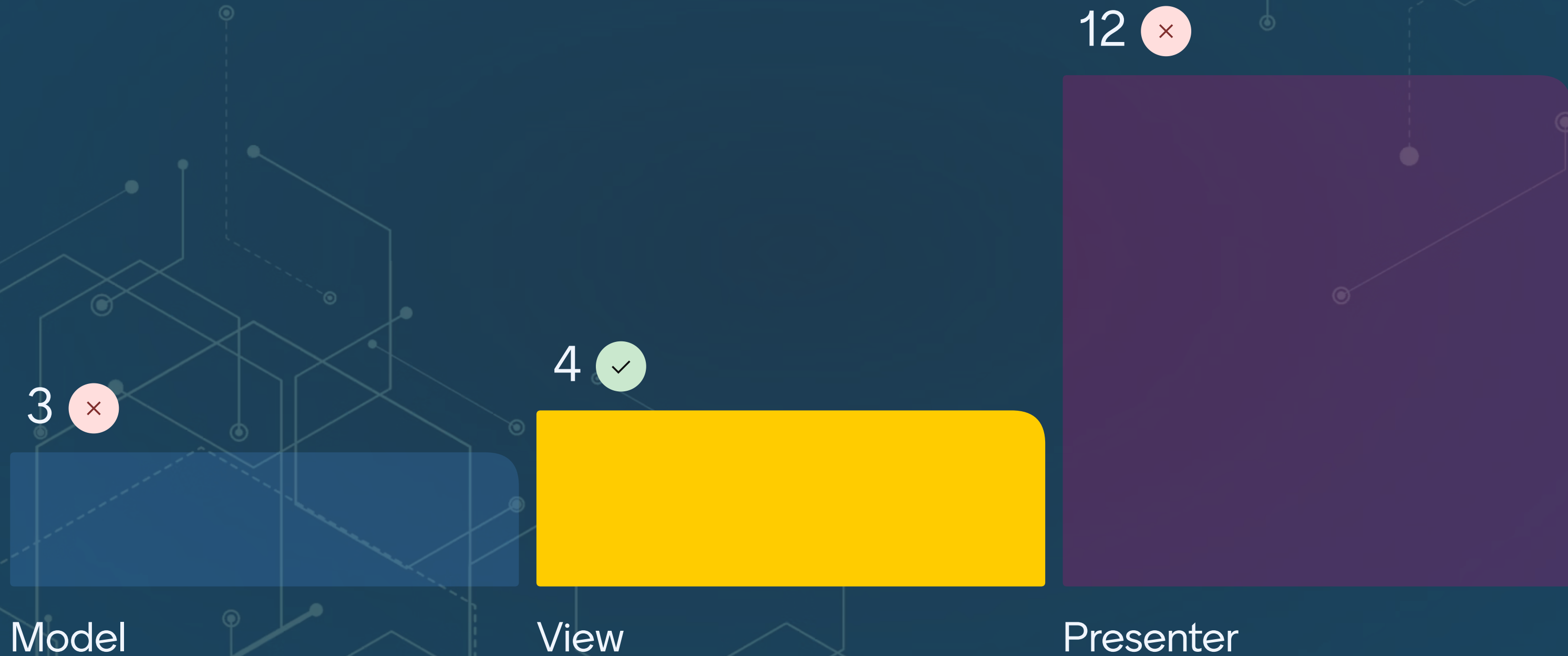
What is the difference between Architectural pattern and Design patterns?

scope and level of
abstraction

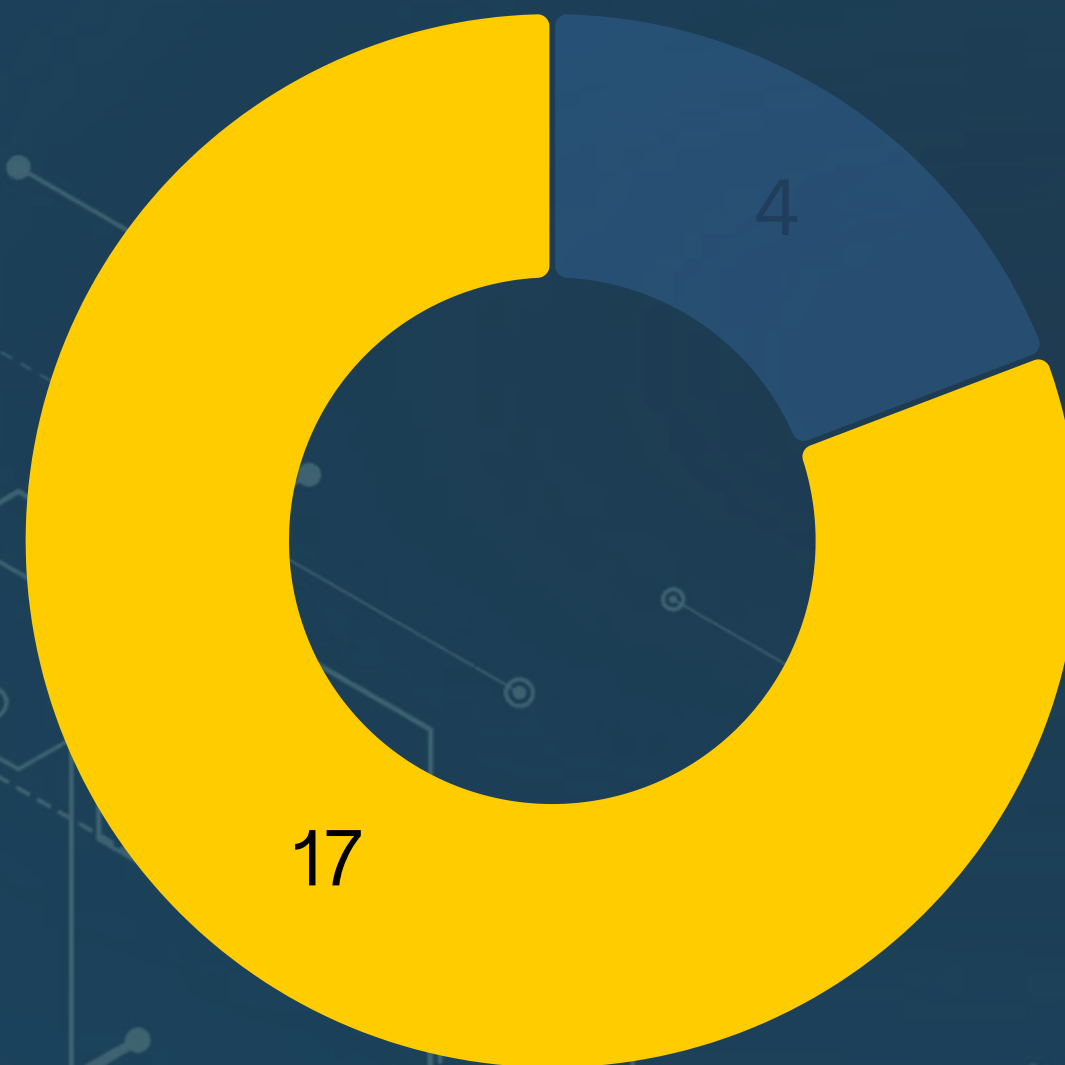
architectural is overall
structure and design is
specific implementation

archit pattern is about
structure of project design
pattern is about code
structure

Who is receiving requests in MVP?

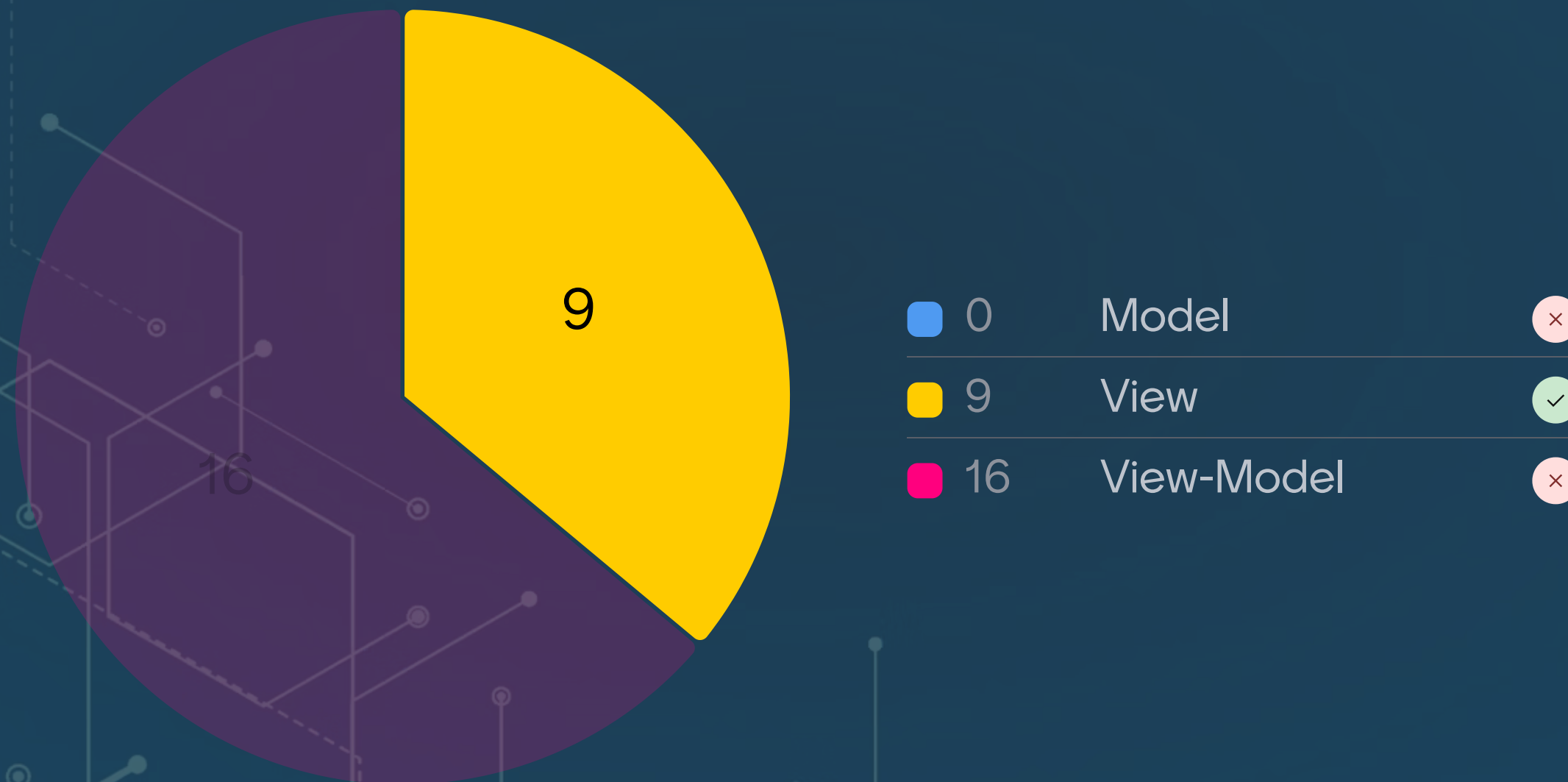


How many *Views* can be referred to *Presenter* in MVP?



4	Multiple	×
17	1	✓
0	2	×

Who is receiving requests in MVVM?



Difference between MVC and MVVM ???

number of views

MVC uses a Controller to handle input and coordinate the Model and View. MVVM uses a ViewModel between the View and Model

well ... i am totally agree with them

I don't know

MVC: 1. for small side projects 2. easy to implement

to handle user input MVC uses Controller, MVVM uses VM + Data Binding

mvc - view knows about controller, mvvm - view model does not know about a view

MVC = controller, MVVM = view

Difference between MVC and MVVM ???

MVC, the Controller receives requests/inputs directly. In MVVM, the View receives inputs.

MVVM: ViewModel connects the View and Model, usually through data binding. MVC: Controller handles user input and updates the Model/View.

in mvc controller handles requests and in mvvm view handles them. mvvm uses data binding

interaction with view
controller view View
viewmodel

mvc - controller receives requests, handles input, updates model, selects view
view can directly observe/read from model in many implementations relationship many-to-many possible (one Controller

The size of the projects and its implementations

mvvm for mobile development mvc for serverside and desktop programs

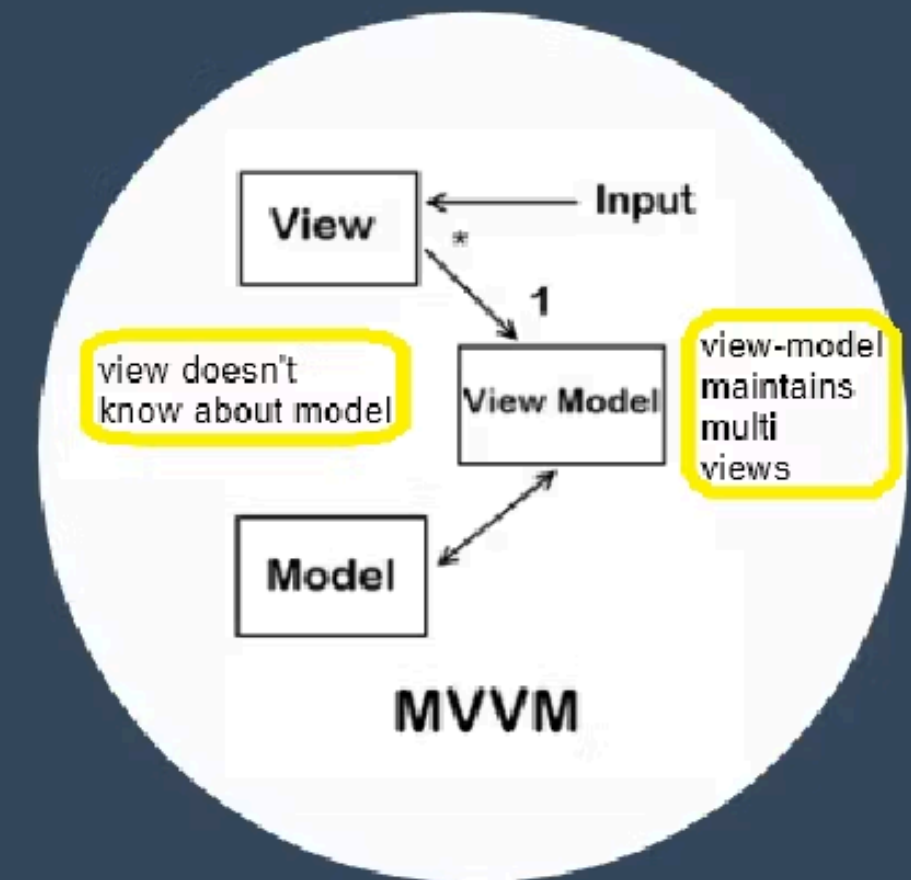
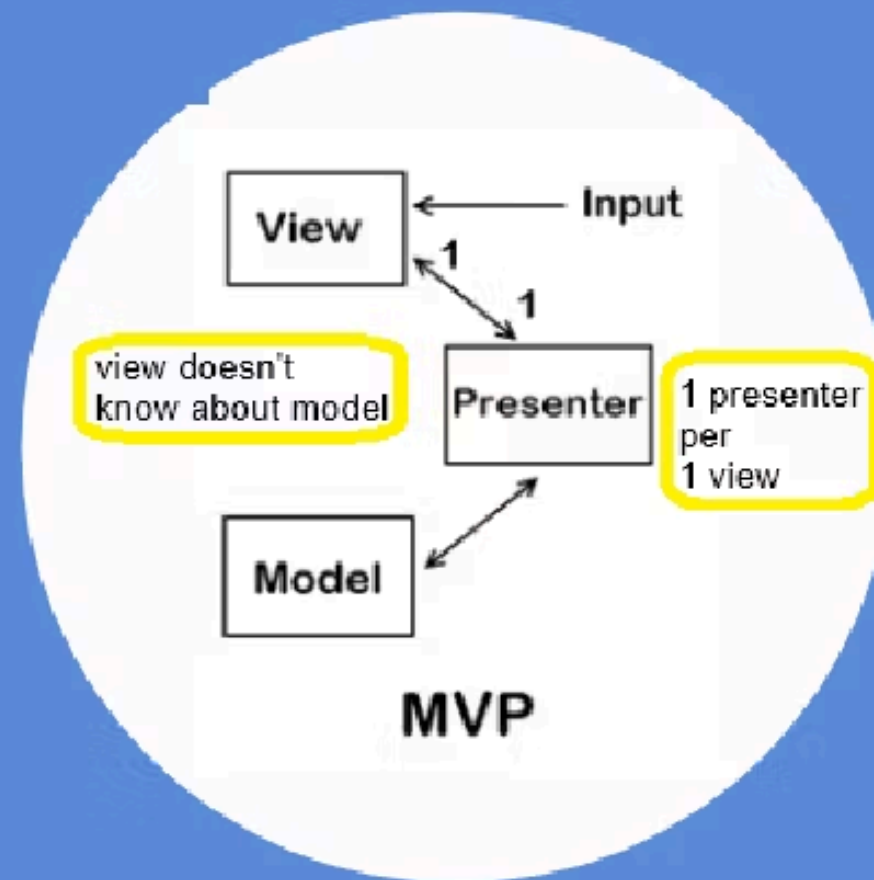
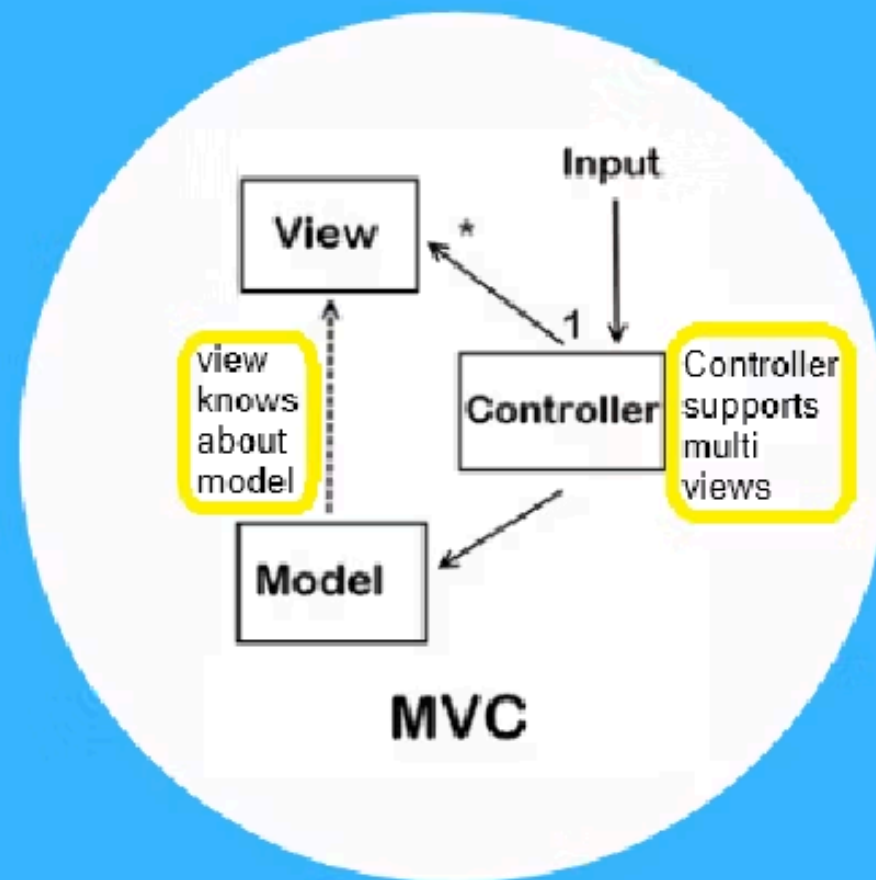
MVC for a server-rendered web app

Difference between MVC and MVVM ???

MVVM: 1. for complex projects 2. easy to implement parallel tasks within team

MVC - View can directly access Model (in some implementations), MVVM - View never directly touches Model — only talks to ViewModel

MVC VS MVP VS MVVM



Simple algorithm to define Pattern

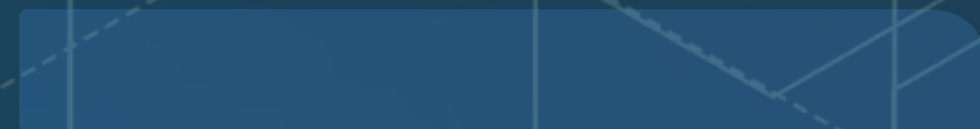
- Notice the request operator...
- If request comes to 'Dispatcher' - it's a Controller => MVC
- If some View layer receive requests => MVP or MVVM
- Notice View relationship...
- If 'Dispatcher' serves only for one View - it's a Presenter => MVP
- If 'Dispatcher' serve for several Views - it's a View-Model => MVVM

What are the layers in *3-layered architecture* ?

18 ✓



2 ✗



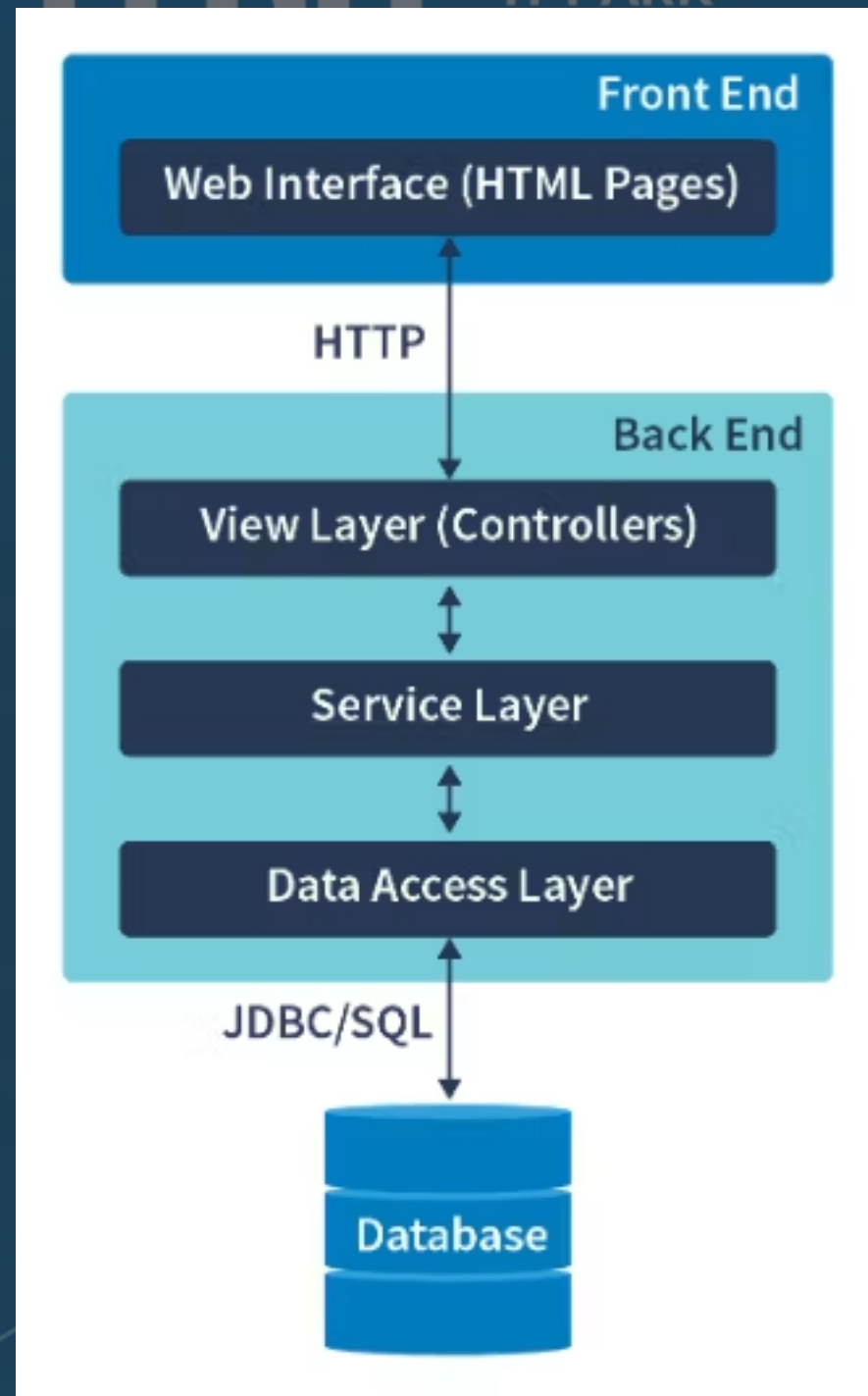
Model, View, Controller

0 ✗



HTML, Java, Database

Controller, Service, DAO



explore an example from ws-02

Architecture Advanced (ws-03)

Code

If you want to follow the examples -
Select an appropriate workshop (ws-...) at

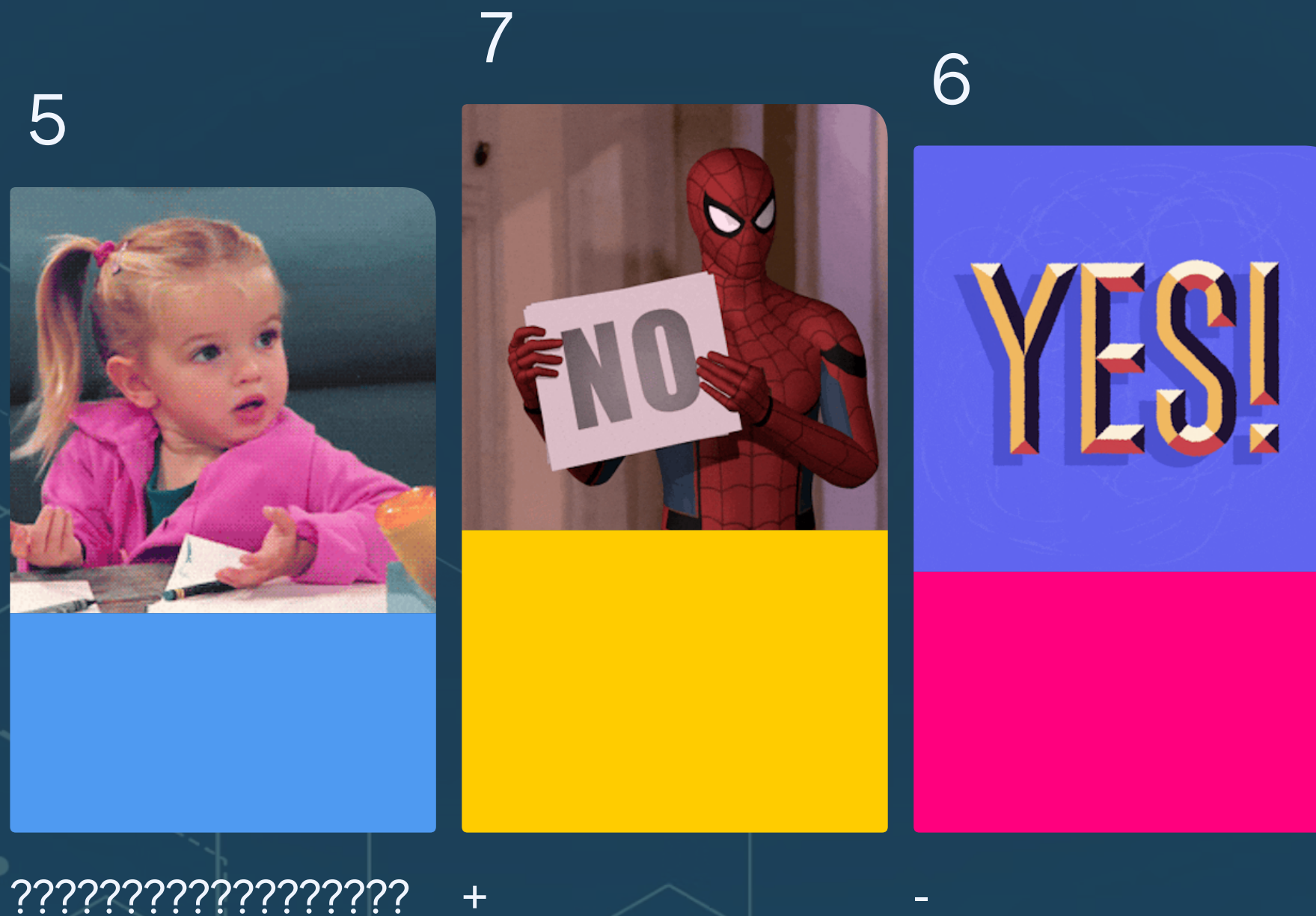
<https://github.com/AndreiRohau/itpu-web-development-with-java-2026>

ERD

Entity Relationship Diagram



Have you started the theory of 'Design of a Web App Database'?



The Entity Relationship (ER) Diagram
is a powerful visual tool for designing relational
database structures.

Explore code from ws-03

<https://github.com/AndreiRohau/itpu-java-web-2025/tree/main/ws-03-architecture-advanced>

Few more

- primary key
- foreign key
- constraint

Primary Key Options

- **Single-Column Simple Key**

Uses one existing natural attribute that is guaranteed to be unique.

- *Example:* SSN in a User table, or ISBN in a Book table.

- **Composite Key (Combined Key)**

Combines two or more attributes together to form a unique identifier when no single attribute is unique on its own.

- *Example:* (CourseID, Semester) in a ClassSchedule table.

- *Example:* (OrderID, ProductID) in an OrderItem associative table.

- ...

...

- **Surrogate Key (Synthetic Key)**

An system-generated unique identifier that has no business meaning, added purely to act as a clean primary key.

- *Example:* Auto-incrementing `id` (`BIGINT`), or a generated `UUID` (e.g., `a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11`).

- **Compound Key**

A specific type of composite key where at least one of the combined fields is a foreign key referencing another entity (common for weak entities).

- *Example:* (`EmployeeID`, `DependentName`) where `EmployeeID` is a foreign key pointing to the `Employee` table.

- **Natural Key**

A real-world attribute inherent to the entity that serves as a unique key.

- *Example:* `EmailAddress` in an `Account` table, or `VIN` (Vehicle Identification Number) in a `Car` table.

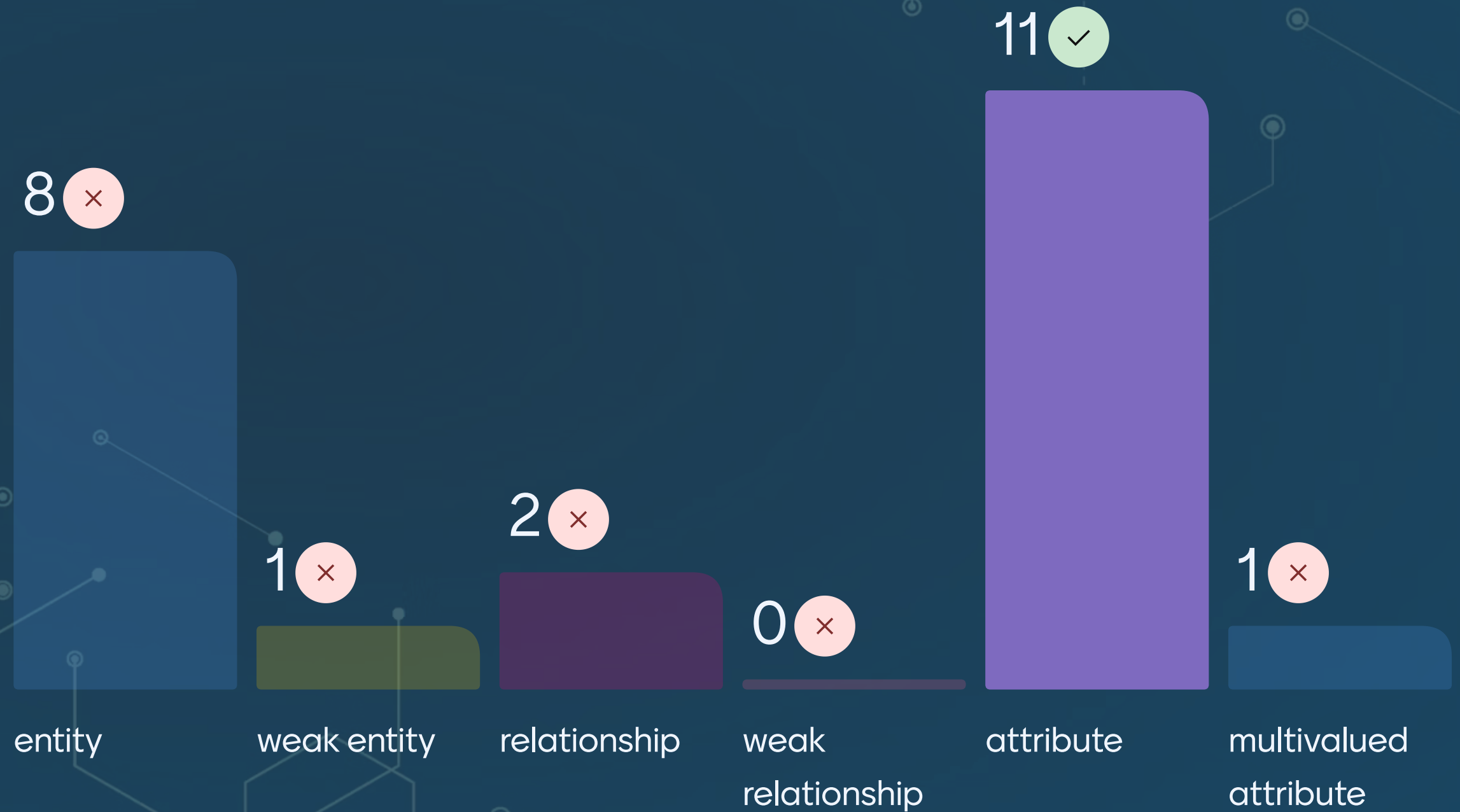
Database normalization

https://en.wikipedia.org/wiki/Database_normalization

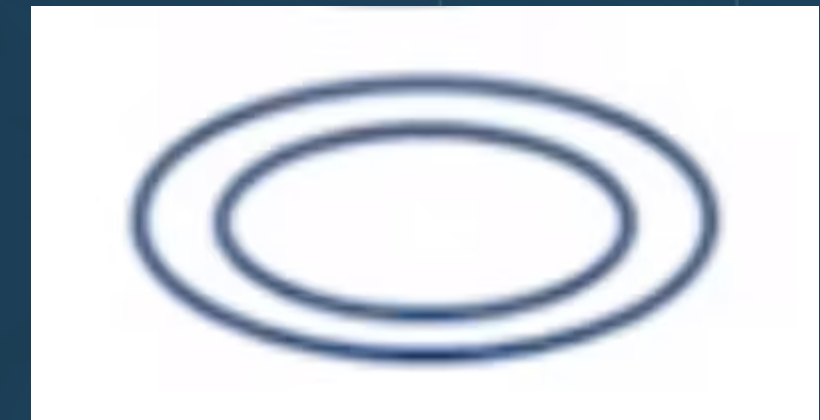
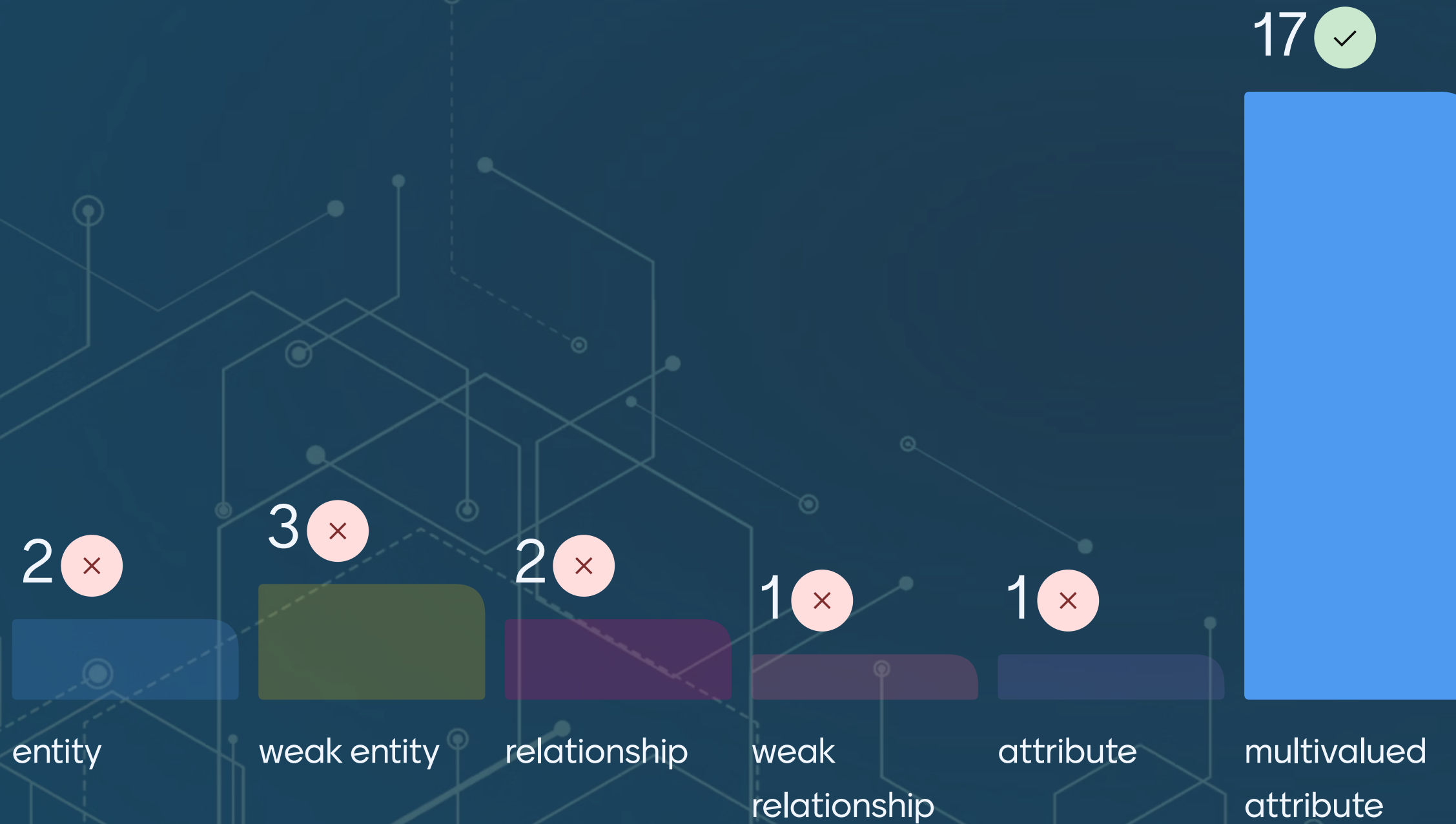
Quiz after a coffee break

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select correct meaning:

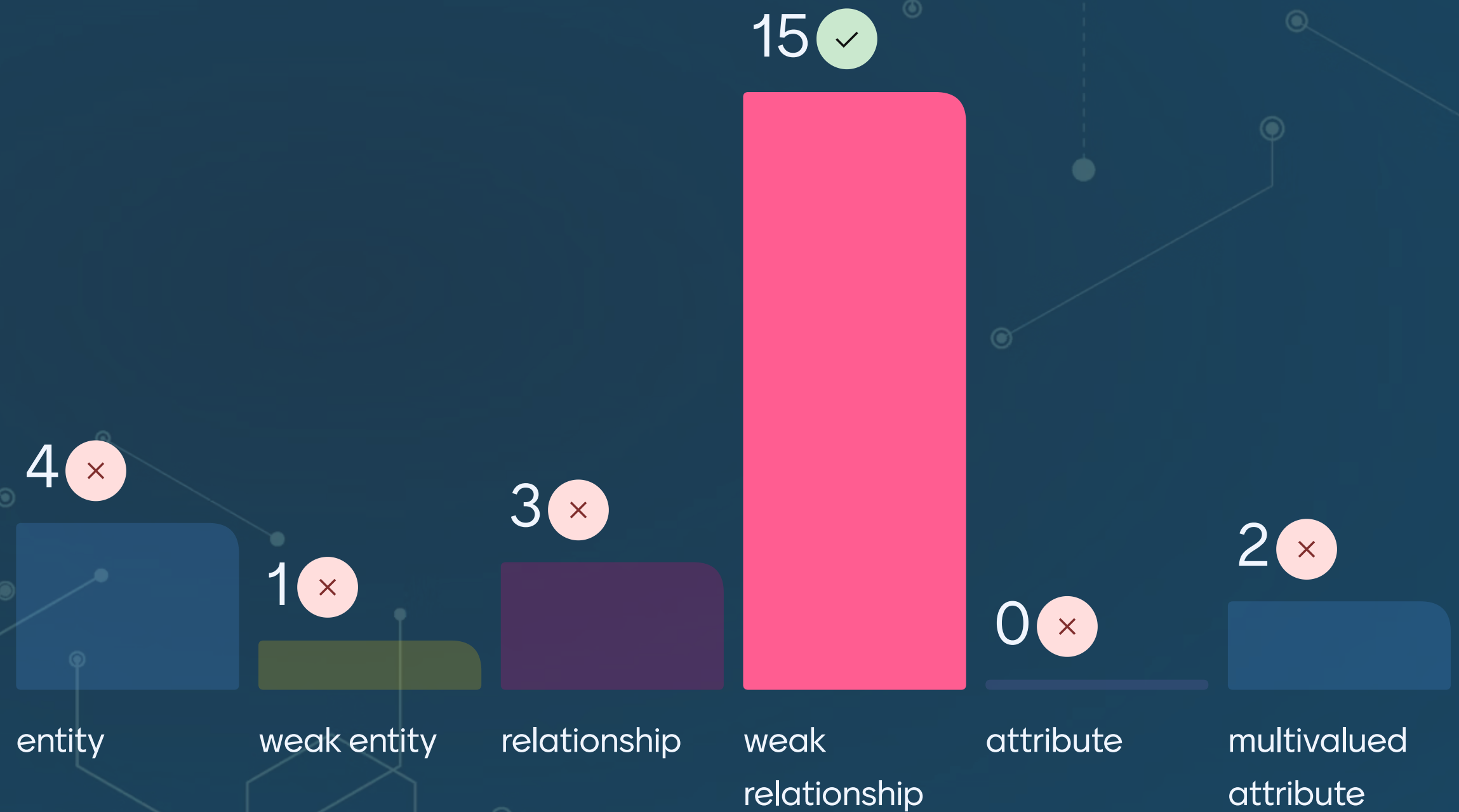
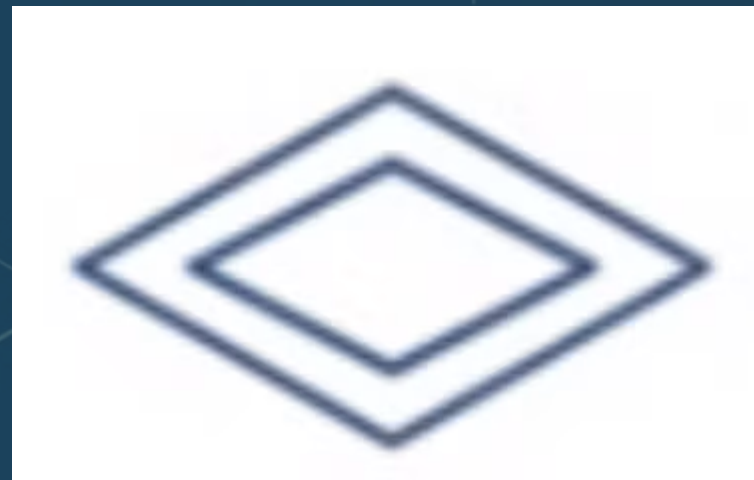


select correct meaning:



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select correct meaning:



Derived Attribute -

21 ✓



calculated from other attribute

3 ✗



got from another entity

3 ✗



got from another relationship

What is the entity attribute that uniquely identifies an item in a table?

26 ✓

1 ✗

Secondary key

1 ✗

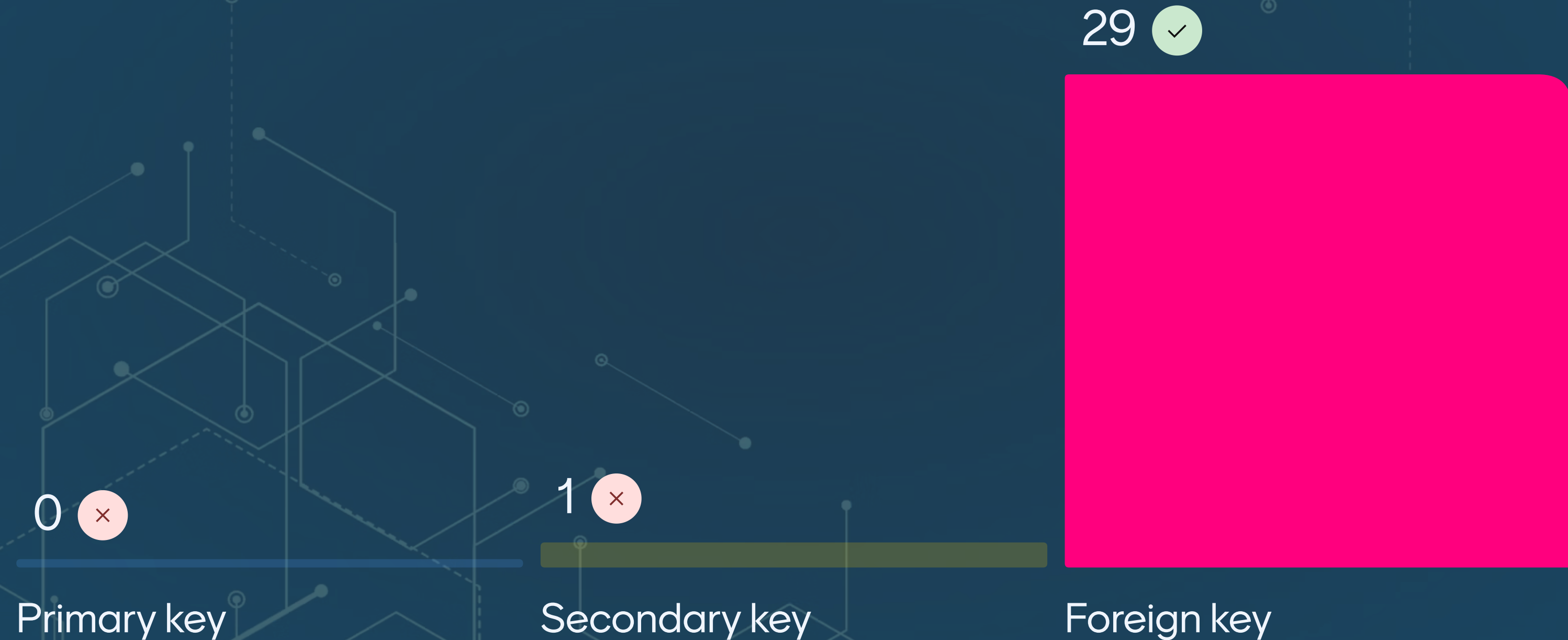
Foreign key

0 ✗

Compound key

Primary key

A key that is used to reference a primary key from another table is



What is used when two or more primary keys from another entities are necessary to determine a unique entity?

13 ✓



Compound key

17 ✗



Composite key

0 ✗

Relational key

What is used when two or more attributes are necessary to determine uniqueness of an entity?

25 ✓



3 ✗



Compound key

0 ✗

Relational key

Composite key

What attribute of a weak entity is used together with primary key of a strong entity to generate primary key for this weak entity?

17 ✓

5 ✗

Compound key

5 ✗

Composite key

Relational key

4



2



Which is a valid relationship between entities?

26 ✓

2 ✗

one too many

0 ✗

one two three

one to one

0 ✗

many to noone

select all true:

20 ✓



mandatory many

3 ✗



many to one

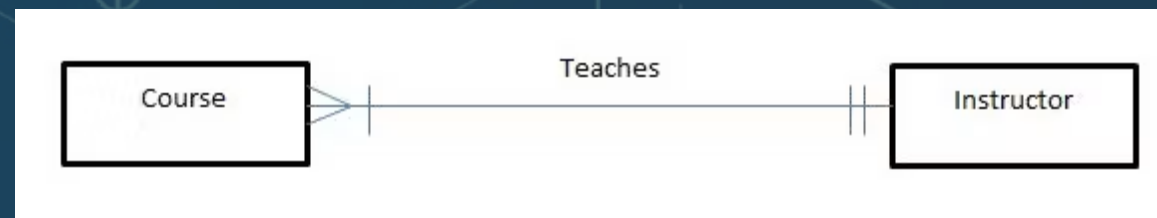
4 ✗



optional many



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select all true:



20 ✓

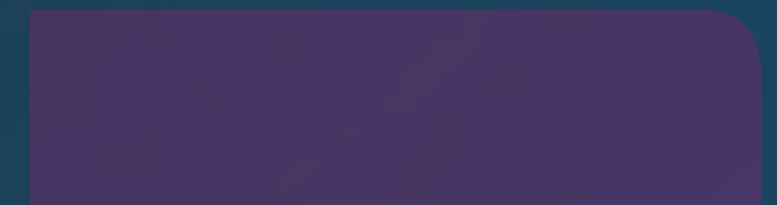


3 ✗



(0,N) to (0,1)

4 ✗

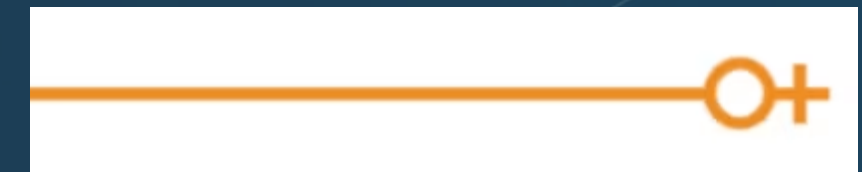


(1,N) to (0,1)

(1,N) to (1,1)

select all true:

24 ✓



2 ✗

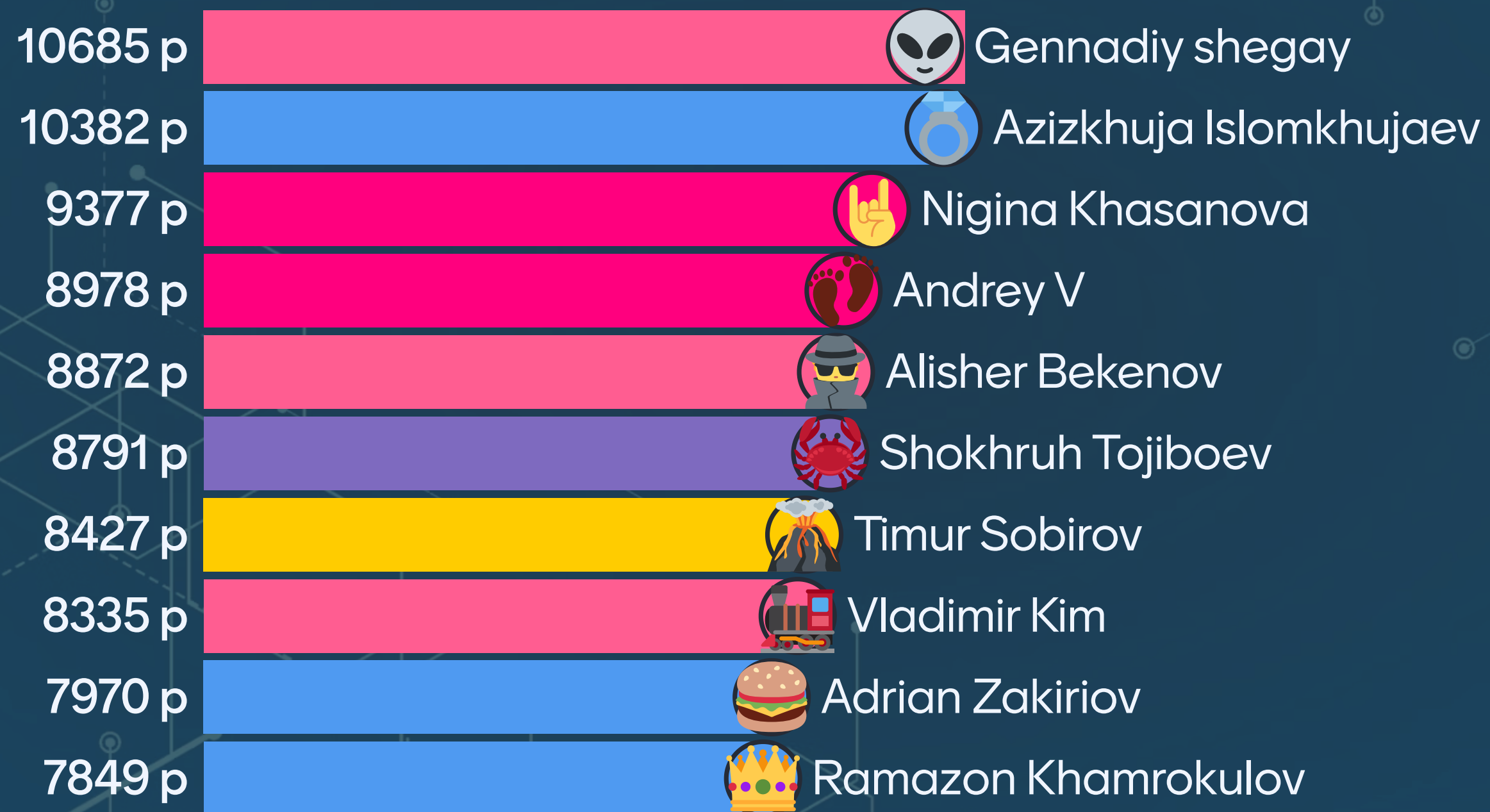
exactly one

1 ✗

one

zero or one

Quiz leaderboard



How to Create an ERD

- Entity
- Relationship
- Cardinality
- Attributes
- follow example: <https://www.guru99.com/er-diagram-tutorial-dbms.html>

Discuss the task

[https://elearn.university.epam.com/courses/
course-
v1:UNIVERSITY+SE124_T3+1223/coursewar
e/044040a3ecdf47bf965758eac25d3e57/5
2a998971fe740988bbf58e0e9029685/](https://elearn.university.epam.com/courses/course-v1:UNIVERSITY+SE124_T3+1223/courseware/044040a3ecdf47bf965758eac25d3e57/52a998971fe740988bbf58e0e9029685/)

Next time: User Interface Design

Questions?

Is one page enough?

